HANDWRITTEN STUDY NOTES

# GEOSYNTHETICS AND REINFORCED EARTH STRUCTURES

**Subject Code : 23A01702a**

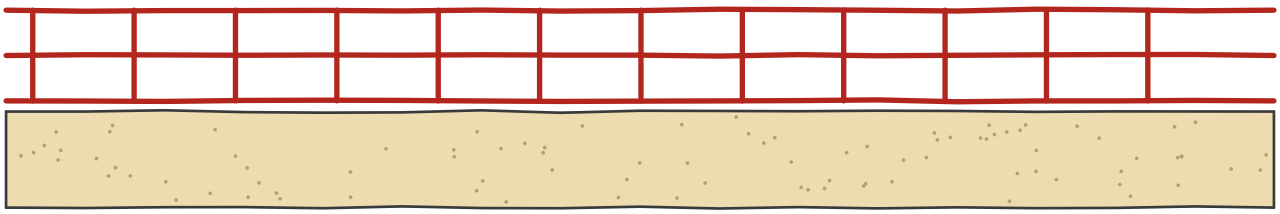
# UNIT – II

## Geosynthetics

*Syllabus topics covered in these notes:*

- Geosynthetics: advantages over conventional materials
- Classification based on material type and function; types of geosynthetics
- Functions of geosynthetics; tests on geosynthetics
- Durability aspects of geosynthetics; applications of geosynthetics

soil + reinforcement = strong composite



# 1. Geosynthetics – Introduction

**Geosynthetics** = geo (earth) + synthetic (man-made polymer). They are **planar, polymeric (synthetic or natural) products** manufactured for use with soil, rock or other geotechnical material as an integral part of a man-made project, structure or system (ASTM D4439).

## Raw materials (polymers)

- **Polypropylene (PP)** – most geotextiles, geogrids; cheap, UV-sensitive, creeps.
- **Polyethylene (HDPE, LDPE, LLDPE)** – geomembranes, geogrids, geonets; chemically resistant.
- **Polyester (PET)** – high strength, low creep; geogrids, woven geotextiles; sensitive to high pH (hydrolysis).
- **Polyvinyl chloride (PVC)** – flexible geomembranes; **Polyamide (nylon)**, **aramid**, **PVA** – special reinforcement, **EPS** – geofoam.
- Additives: carbon black (2–3 %), antioxidants and UV stabilisers are added to improve durability.

## Advantages over conventional materials

| Conventional (sand, stone, steel, clay) | Geosynthetics |
|---|---|
| Natural resources, variable quality | Factory made – uniform, quality controlled |
| Thick layers, heavy haulage | Thin rolls, easy transport and handling |
| Slow, labour and equipment intensive | Fast installation, little equipment |
| Steel corrodes; clay layers crack | Corrosion free, flexible, chemically resistant |
| Large quarrying; high carbon cost | Saves aggregate and land; eco-friendly |
| High cost for long haul | Cost saving 20–60 % in many applications |

## Limitations

- Damaged by **UV light** if left exposed; sensitive to fire and some chemicals.
- **Creep** under sustained load; installation damage; long-term data limited (≈ 60 years of use).

- Needs correct design, specification and good site supervision.

## 2. Classification of Geosynthetics

Geosynthetics can be classified (i) by **type / product form**, (ii) by **polymer (material)** and (iii) by **function**. Main types are shown below.

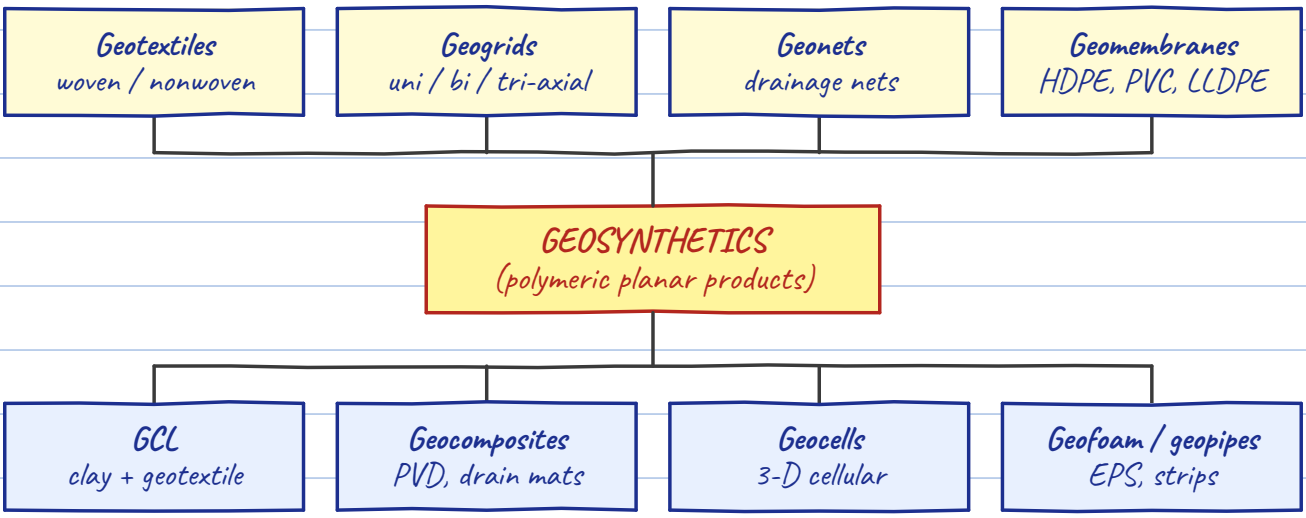



Fig 2.1 Types of geosynthetics

### 2.1 Types of geosynthetics

(i) **Geotextiles** – permeable, fabric-like sheets of PP/PET/PE fibres.

- **Woven** – slit-film or monofilament yarns interlaced at right angles; high strength, low strain, low permeability → reinforcement, separation.
- **Nonwoven** – fibres bonded randomly (needle-punched, heat-bonded, resin-bonded); thick, high elongation, high permeability → filtration, drainage, protection.
- **Knitted** – loops formed, used for special reinforcement and erosion control.

(ii) **Geogrids** – open grid-like sheets with large apertures (10–100 mm), stiff ribs; interlock with aggregate → **reinforcement**.

- **Uniaxial** (high strength in one direction – walls, slopes), **biaxial** (both directions – roads, platforms), **triaxial** (triangular apertures – better load spread).

- Manufactured by extrusion & stretching (HDPE/PP), weaving + coating (PET) or welding of strips.

**(iii) Geonets** – two sets of parallel ribs crossing at an angle; high in-plane flow → **drainage.**

**(iv) Geomembranes** – continuous impermeable sheets (HDPE, LLDPE, PVC, fPP, EPDM); $k \approx 10^{-11}$ to $10^{-13}$ cm/s → **barrier / containment**, thickness 0.5–2.5 mm.

**(v) Geosynthetic clay liners (GCL)** – sodium bentonite sandwiched between geotextiles; self-healing; $k \leq 10^{-9}$ m/s.

**(vi) Geocomposites** – combination of two or more types, e.g. geotextile + geonet drain, **prefabricated vertical drains (PVD / wick drains)**, geogrid + nonwoven.

**(vii) Geocells (geoweb)** – 3-D honeycomb cells filled with soil/aggregate → confinement, slope and base stabilisation.

**(viii) Others** – geofoam (EPS blocks, light fill), geopipes, geostrips/geobelts, geomats, erosion control blankets, geotubes / geobags.

## 2.2 Classification by polymer and by function

| Product | Common polymer | Main function |
|---|---|---|
| Woven geotextile | PP, PET | Separation, reinforcement |
| Nonwoven geotextile | PP, PET | Filtration, drainage, protection |
| Geogrid | HDPE, PP, PET | Reinforcement |
| Geonet | HDPE | Drainage |
| Geomembrane | HDPE, PVC, LLDPE | Containment (barrier) |
| GCL | Bentonite + PP | Containment |
| Geocomposite / PVD | PP + PE core | Drainage, filtration |
| Geocell | HDPE | Reinforcement, confinement |

## Manufacturing in brief

- **Woven GT:** polymer → yarns (slit-film / monofilament) → woven on looms.

- **Nonwoven GT:** filaments extruded → random web → bonded by **needle punching** (barbed needles entangle fibres), **heat bonding** or resin.
- **Geogrid:** extruded sheet punched with holes and **stretched** (uni- or biaxially) to align polymer chains; or PET yarns knitted/woven and PVC/bitumen coated.
- **Geomembrane:** flat-die / blown-film extrusion or calendering; panels joined at site by **thermal (hot wedge) fusion** or **extrusion welding**; seams tested by peel, shear and air-channel tests.
- **GCL:** bentonite layer needle-punched or stitched between two geotextiles; swells on hydration and seals punctures.

## Prefabricated vertical drain (PVD) – geocomposite

Band drain about 100 mm × 4 mm: plastic **grooved core** wrapped by a **filter geotextile**, pushed into soft clay with a mandrel at 1–2.5 m spacing. It shortens the drainage path from the clay thickness to **half the drain spacing**, so consolidation takes months instead of years.

$$t = (D^2 / 8c_h) \cdot [\ln(D/d_w) - 0.75] \cdot \ln[1/(1 - U)]$$

$D = 1.05\ s$ (triangular) or $1.13\ s$ (square pattern), $d_w = 2(a + b)/\pi$ = equivalent drain diameter, $c_h$ = horizontal coefficient of consolidation, $U$ = degree of consolidation (Barron; smear ignored).

### Example (PVD)

$s = 1.5$ m triangular → $D = 1.575$ m; $a = 100$ mm, $b = 4$ mm → $d_w = 66$ mm; $c_h = 3\ m^2/yr$; $U = 90\ \%$

$t = (1.575^2/(8\times3)) \times [\ln(23.9) - 0.75] \times \ln(10) = 0.1034 \times 2.42 \times 2.303 =$ 0.58 year (≈ 7 months)

# 3. Functions of Geosynthetics

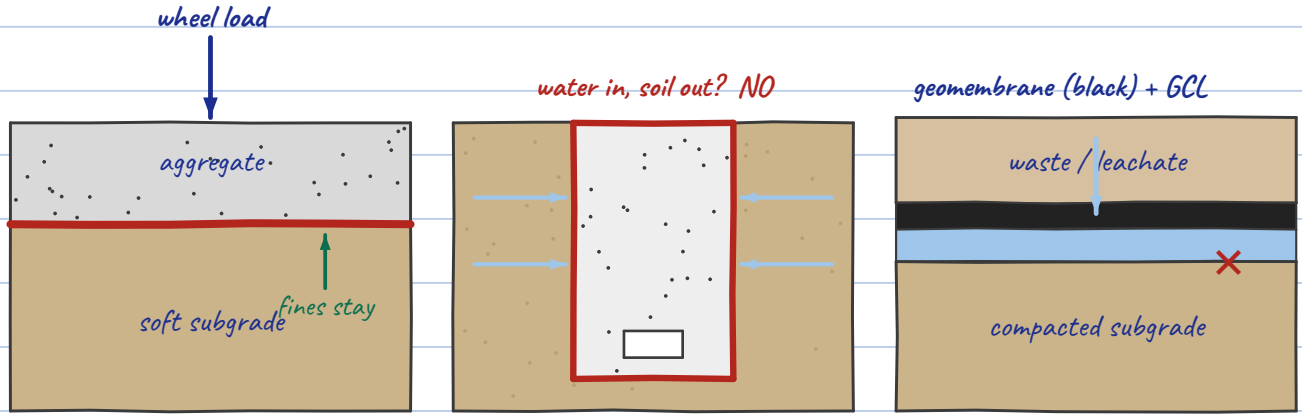



Fig 2.2 Separation, filtration/drainage and containment

**1. Separation** – keeps two dissimilar materials (soft subgrade and aggregate base) from mixing. Without it, fines pump up and stone punches down, so the base thickness is lost. Properties: tensile strength, puncture, burst, apparent opening size. Saves aggregate (≈ 30–50 %).

**2. Reinforcement** – adds **tensile resistance** to soil (geogrid, woven geotextile, geostrip): retaining walls, steep slopes, embankments on soft soil, bearing capacity improvement. Properties: tensile strength and modulus, creep, soil-interaction (pull-out).

- **Lateral restraint** – tensile reinforcement holds soil/aggregate from spreading sideways (see confining pressure concept).
- **Bearing-capacity (failure-plane) mechanism** – forces the failure surface to a deeper, stronger zone.
- **Tensioned-membrane effect** – after rutting, curved sheet carries part of the wheel load by vertical component of tension.

**3. Filtration** – allows water to pass but retains soil particles (like a sand filter). Typical use: behind retaining walls, around drains, under riprap.

### Filter criteria (geotextile)

Retention: $O_{95} \le B \cdot D_{85}$ ($B = 1$ for $C_u \le 2$ or $\ge 8$; $B = 0.5\,C_u$ for $2 < C_u < 4$; $B = 8/C_u$ for $4 \le C_u \le 8$)

Permeability: $k_g \ge k_s$ (critical: $\ge 10\,k_s$); permittivity $\psi \ge 0.5\ s^{-1}$ (fines < 15 %), 0.2 (15–50 %), 0.1 (> 50 %)

Anti-clogging: $O_{95}$ not too small; nonwoven preferred for fine soils.

**4. Drainage (in-plane flow)** – carries water/gas within its plane (geonet, thick nonwoven, geocomposite). Transmissivity $\theta = k_p \cdot t$ ($m^2/s$); flow $q = k_p\, i\, t\, W$. Used in landfill leachate collection, behind walls, under roads, PVDs for consolidation of soft clay.

**5. Containment (barrier)** – geomembrane / GCL stop liquid or gas movement: landfill liners and caps, canal and pond lining, mining heap-leach pads.

**6. Protection** – cushion layer (thick nonwoven) protects a geomembrane from puncture by stones; **7. Erosion control** – mats and blankets hold soil on slopes and channels until vegetation grows.

### Example 1

Example (filter): soil $D_{85} = 0.4$ mm, $C_u = 5 \rightarrow B = 8/5 = 1.6$

$O_{95} \le 1.6 \times 0.4 = 0.64$ mm (geotextile AOS must be smaller than this).

## 4. Tests on Geosynthetics

Properties are tested in the laboratory as per **ASTM / ISO / IS 13162** (Indian standard series for geotextiles).

### 4.1 Physical properties

- **Mass per unit area** ($g/m^2$) – ASTM D5261; **thickness** (mm) at 2 kPa – ASTM D5199; **specific gravity / density** of polymer; **roll size**; **stiffness**.

### 4.2 Mechanical properties

- **Wide-width tensile test** (ASTM D4595 / ISO 10319): specimen 200 mm wide × 100 mm gauge, strain rate 10 %/min. Gives $T_{ult}$ (kN/m), strain at failure, secant modulus $J = T/\varepsilon$ (at 2 %, 5 %).

- **Grab / strip tensile** (ASTM D4632) – index test, small specimen held by 25 mm jaws.
- **Trapezoidal tear** (D4533), **puncture** (CBR puncture D6241, drop-cone), **burst** (D3786).
- **Creep test** (D5262, tension creep) – constant load held for 1000–10 000 h.
- **Seam / joint strength**; **interface friction** (direct shear, pull-out, inclined plane – see Unit I).

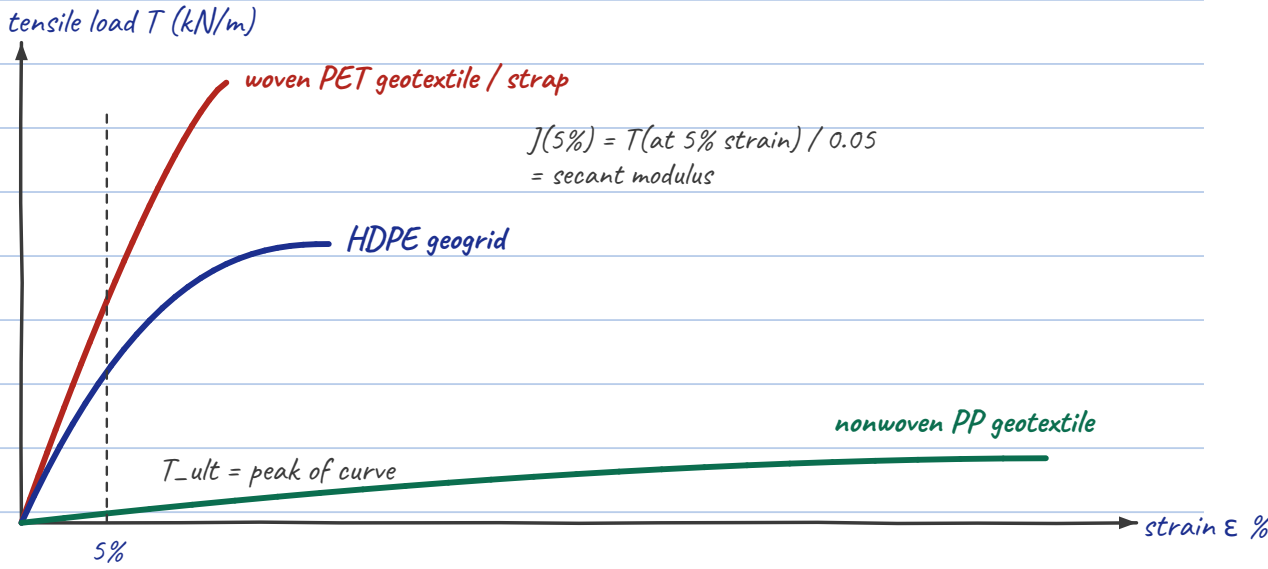



Fig 2.3 Load–strain behaviour of geosynthetics

## 4.3 Hydraulic properties

- **Permittivity / permeability** (ASTM D4491; constant or falling head) – cross-plane flow.
- **Apparent opening size, AOS ($O_{95}$)** (ASTM D4751) – dry sieving with glass beads.
- **Transmissivity** (ASTM D4716) – in-plane flow under load and gradient.
- **Gradient ratio / clogging** (ASTM D5101) – long-term filter compatibility with soil.

## Examples

Wide-width test: specimen 200 mm wide; peak load 12 kN → $T_{ult}$ = 12/0.2 = 60 kN/m.

Load at 5 % strain = 6 kN → $T_{5\%}$ = 30 kN/m → $J$ = 30/0.05 = 600 kN/m.

Permittivity $\psi = Q / (A \cdot \Delta h)$ $[s^{-1}]$; $k_n = \psi \cdot t$. If $\psi = 0.8\ s^{-1}$, $t$ = 2 mm → $k$ = 0.8 × 0.002 = $1.6 \times 10^{-3}$ m/s.

## 4.4 Endurance and durability tests

- **UV resistance** (ASTM D4355) – xenon-arc weatherometer, 500 h; retained strength reported.
- **Chemical, oxidation and hydrolysis** (immersion in solution at high temperature), **biological** resistance (soil burial).
- **Installation damage** (ASTM D5818) – field compaction trial with aggregate, then retained strength.
- **Abrasion** (D4886), **temperature** effects, **stress cracking** (geomembranes, D5397).

## Summary of tests

| Property | Test (ASTM) | Result / unit |
|---|---|---|
| Mass per unit area | D5261 | $g/m^2$ |
| Thickness | D5199 | mm at 2 kPa |
| Tensile (wide width) | D4595 | kN/m, % strain, $J$ |
| Puncture / burst | D6241 / D3786 | N / kPa |
| Permittivity | D4491 | $s^{-1}$ |
| Apparent opening size | D4751 | O95 (mm) |
| Transmissivity | D4716 | $m^2/s$ |
| UV resistance | D4355 | % strength retained |

## 5. Durability Aspects

Design life of geosynthetics (60–120 years for permanent works) depends on resistance to the following degradation mechanisms:

| Mechanism | Material mostly affected | Remedy |
|---|---|---|
| Creep / stress rupture | PP, HDPE (PET low) | Use creep factor; choose PET |
| Installation damage | All (esp. nonwoven, thin) | Cover quickly, select fill |
| UV degradation | PP, PE (without carbon black) | Cover within days / weeks |
| Oxidation (thermal) | PP, PE | Antioxidants, cooler soil |
| Hydrolysis | PET at pH > 9 and warm | Check soil pH, use PP/HDPE in lime soil |
| Chemical attack / stress crack | HDPE membranes, PP | Compatibility test |
| Biological, clogging | Filters, drains | Correct AOS, gradient ratio |

### Allowable (long-term design) strength

$$T_{al} = T_{ult} / (RF_{ID} \times RF_{CR} \times RF_{D} \times FS)$$

$RF_{ID}$ = installation damage (1.05–3.0), $RF_{CR}$ = creep (PET 1.6–2.5; PP 4–5; HDPE 2.6–5), $RF_{D}$ = chemical/biological durability (1.1–2.0), FS = overall factor of safety (often incl. uncertainty).

**Example 2**

$T_{ult}$ = 60 kN/m, $RF_{ID}$ = 1.1, $RF_{CR}$ = 2.0, $RF_{D}$ = 1.1, (FS applied separately)

$T_{al}$ = 60 / (1.1 × 2.0 × 1.1) = 60 / 2.42 = 24.8 kN/m

# 6. Applications of Geosynthetics

| Area | Geosynthetic used | Function |
|---|---|---|
| Roads, railways, airfields | Geotextile, geogrid, geocell | Separation, reinforcement |
| Retaining walls, steep slopes | Geogrid, geostrip, woven GT | Reinforcement |
| Embankments on soft soil | High strength GT / geogrid, PVD | Reinforcement, drainage |
| Landfills, ponds, canals | Geomembrane, GCL, geonet | Containment, drainage |
| Drains, dams, subsurface | Nonwoven GT, geocomposite | Filtration, drainage |
| Coastal / river works | GT tubes, mats, filters | Erosion control, filtration |
| Tunnels, mining | Membrane, shotcrete liner, GT | Waterproofing, protection |

## Typical landfill liner system (bottom to top)

1. Compacted subgrade → **compacted clay / GCL** (primary barrier).
2. **HDPE geomembrane** (1.5–2 mm) – leachate barrier.
3. **Nonwoven geotextile cushion** – protects the membrane from stones.
4. **Drainage layer** (gravel or geonet) with perforated leachate pipes.
5. **Filter geotextile** – keeps waste fines out of the drainage layer, then waste.

## Selection of a geosynthetic

- Identify the **main function** first (separation, reinforcement, filtration, drainage, containment).
- Check required **properties** (strength, AOS, permittivity) against site soil and loads.
- Check **durability** (pH, temperature, UV, chemicals), **installation survivability** and **cost**.

## Important questions

1. Define geosynthetics. List their advantages over conventional materials.
2. Classify geosynthetics by type, material and function; explain geotextiles and geogrids.

3. Explain the functions of geosynthetics with sketches and examples.
4. Describe the wide-width tensile test and permittivity test.
5. Discuss the durability aspects of geosynthetics; explain reduction factors.